

Framework Review — Cross-Manifesto Analysis and the Temporal Extension

FrameworkReview mission: peer merge (TuringAward, MINSKY, FORMALIS) + Temporal Commit

21 July 2026

Abstract

This document assembles the FrameworkReview mission in two parts. **Part I** is a merged systemic review of the Ledger Constitution together with the Market Data and Valuation Manifestos, read as *one composed system* by three independent reviewers (TuringAward on architecture, MINSKY on type discipline, FORMALIS on cross-manifesto coherence); findings are attributed and divergences kept as positions, never averaged. **Part II** is the unanimous round-11 consensus of the Temporal Committee: a substrate proposal placing the whole framework — market data, valuation, and lifecycle — on one durable-execution machine that folds observations and re-entered observations through the one door, with the immutable log remaining the sole book of record. Both parts turn on one boundary — where a deterministic function of the record becomes a stored artifact: Part I finds the gravest open collision there; Part II contains it by construction.

Part I — Merged Systemic Review

Three reviewers read the three manifestos independently as one system (TA: architecture/algorithms; MK: type discipline, illegal states; FL: cross-manifesto formal coherence). Findings are attributed; divergences are kept as positions, never averaged. Severity CRITICAL/HIGH/MED/LOW — TA's MAJOR→HIGH, MINOR→LOW.

1. Systemic verdict

The three documents compose *without a vicious logical cycle*: definitions and discharges each flow one way [FL]. But they share one fault line — the boundary at which a deterministic function of the record becomes a *stored* artifact — and all three reviewers, apart, ranked the collision there the gravest finding. A **version-layering inversion** compounds it: the parent clause a child leans on (MD-16, Market Data 1.3) postdates and is uncited by that child (Valuation 1.0, subordinate to Market Data 1.1) [FL]. The recurring defect beyond the headline is **drift-by-relabeling at the seams between authors and versions** — a guarantee kept reading true by renaming its residue (“event-outcome,” “forbidden,” “ $\mathcal{D} \equiv \text{VM-5}$,” “structurally impossible”). Correctness holds where one author states a thing once (VM-8 extends MD-14 without re-proving; MD-13 defers the sandwich to VM-9); it frays where documents meet. These are composition gaps and relabelings, not thesis errors.

2. Headline finding (unanimous, gravest)

H1 [all three — gravest]. A ledger-computable number is stored on one side of the framework and parked as unstorable on the other. MD-16 records a gate decision’s *computed functionals and percentiles* — each a map from a derived state m^* to a number, a *projection* by C-4.11’s own definition — as stored fields, while the Valuation Manifesto *parks* the

identical act (storing σ_{prod}^2 , PARK-1) because “a projection stores nothing.” MD-16 declares “no collision with C-4.11 ... an event-outcome, not a projection.” Three angles converged: [TA] a *certified* article (Market Data 1.3) silently depends on an unresolved sibling park — rule C-4.11 literal and it breaks retroactively; [MK] “event-outcome” *is* the reclassification PARK-1 refused without the owner, the CLAUDE.md §1 worse failure (a guarantee kept true by renaming its residue, eroding the reconciliation root C-1.2–1.3); [FL] silent non-conformance with a blast radius the parking index hides — and, surgically, recording the pass/fail *bit* is fine, only the numeric *fields* materialise a projection. *Divergence preserved (fix)*: FL’s remedy needs *no* amendment — ground the recording on MD-11/C-14.15 (a simulated path must capture its outputs) and drop “not a projection”; else fold the fields under PARK-1. TA/MK treat the whole recording as materialisation, FL only the fields. *Complementary [TA]*: C-2.8 (“the seed ... every path replays exactly”) itself overclaims against MD-11 (“a path that recorded only the seed could not replay”); amend C-2.8 to match MD-11.

3. Advice — ordered, merged (highest leverage first)

1. [all three] **Rule PARK-1 with *both* documents on the table** — its ruling governs VM PE-6 *and* MD-16’s recorded gate fields; “nothing depends on storage” is true only inside the Valuation Manifesto. Precedes any other ratification.
2. [all three] **Legislate “may a deterministic function of the record be stored?” once** — one clause over C-4.11, PARK-1, MD-16 (PARK-1’s three-condition materialised projection is the candidate), so no document decides it by private nomenclature.
3. [FL, TA] **Freeze the parent text the children consumed, then re-rate** — C-2.7 and C-12.6 (reworded by v1.4/v1.41, both ratification-pending) carry MD-4/MD-5/MD-13/VM-2/VM-9; fix the stale headers (VM 1.0 cites parent files that do not exist).
4. [all three] **Decide prevention-vs-detection at the Constitution** — add a forward-pointer on MD-9 to MD-16’s carve-out; align “forbidden” language to the actual mechanism (detection, recorded-and-flagged).
5. [FL, MK] **Tighten the A5 discharge** — make Gate 1 an interior-membership gate over the move’s neighbourhood, or downgrade “grounds” to “guarantees the endpoint”; wire VM-10’s finite-shift states to MD-16’s gates.
6. [MK] **Complete the typed arrow algebra (C-3.6)** — name the producer of the observation-recording and re-entered-observation arrows, or ratify observable-is-a-unit.
7. [FL, TA] **Add a firing witness to every new invariant** (Gate 1/2, VM-6, VM-7) and state a *liveness* commitment: every stale flag, broken chain, undecidable gate carries a deadline, owner, and fallback.

4. Criticisms — wrong or weakly argued today

- **C1 [MK, HIGH]. The recording arrow has no typed producer.** C-3.6 types only *contract* : (Event, Ledger) $\rightarrow$ Transaction; the observation-recording and re-entered-observation transactions have none — MD-1 attributes recording to the Events Executor, which C-5.4 says “never writes the ledger.” The arrow crossing the trust boundary has unspecified producer, input type, and validation.
- **C2 [all three, HIGH/MED]. Prevention vs detection: two polarities, one unannotated.** MD-9 “by detection, not admission” vs MD-16 Gate 1 “prevention, not detection,” reconciled only inside MD-16 [MK C3, FL C6]. *Preserved*: TA goes further — Gate 1 prevents on an *economic* property *outside* the Transaction Executor, a third

correctness category C-13 never sanctions (HIGH); MK/FL rate it MED and read MD-16 as honest (“a consumer over-trusts”).

- **C3 [FL, HIGH; corr. MK]. Gate 1 discharges VM PE-1(A5) incorrectly.** Gate 1 guarantees one endpoint $m^* \in \Theta_{AF}$; A5 needs $\Sigma(\cdot; S) \in \mathcal{A}$ on a *neighbourhood* with *interiority* — gated at neither end (the origin surface is admissible only by MD-9 detection). *Corr. [MK W5]:* a finite bump (VM-10, spot +20%) falls between the infinitesimal discharge and the finite gate. *Mitigant [FL]:* poles are recorded diagnostics, so nothing breaks silently; only the *discharge claim* is false.
- **C4 [MK, HIGH]. NAV carries two types.** C-8.1/8.2 type NAV as a bare cash number; VM-2 says a valuation without coordinates “is not one” and may be stale/broken — different guarantees in different documents (see W1).
- **C5 [FL, MED-HIGH]. MD-4’s “every observation bears three times” fails for VM’s own observations.** By C-2.7 an emitted event bears no monitor time; a re-entered observation (VM-1: every model-priced leg) is emitted, not observed. Fix: “bears *up to* three times.”
- **C6 [TA, HIGH]. C-2.2 is narrowed in the children without amending the parent.** Unconditional “to the last minor unit” is made conditional by MD-6/VM-2 (re-deriving a model mark “needs the retained model and numerical environment ... outside this scope”); every model-mark’s bit-for-bit claim (VM-8, VM-11) rests on it. C-Auth.3 requires amending the Constitution first.
- **C7 [TA, HIGH]. “Look-ahead is structurally impossible” (MD-4) is not structural.** “By default” concedes a non-default: VM-2’s legal decoupled query (position as-at yesterday, market cut today) inside a VM-11 trajectory reintroduces look-ahead — a discipline to remember. *Adjacent:* VM-11’s “valuation at every step” vs gate-*undecidable* holes [FL C5]; MD-4 fixes the as-of/as-at indexing C-12.1 left to the spec [MK C5].
- **C8 [all three, MED]. The world-map family is asserted, not constrained.** [TA] three names for one concept (*dynamic/shift/D*) stress C-Auth.4; [FL] MD-16’s “VM-5 $\equiv \mathcal{D}$ ” is containment not identity, dropping the correlation channel; [MK] a surface’s $\mathcal{D}$ and its vols’ per-datum dynamics must agree *nowhere*. Also stale subordination: A5’s only discharge (MD-16) postdates and is uncited by VM 1.0 [FL C9].

5. Potential weaknesses — what breaks later, and under what condition

- **W1 [MK, HIGH]. Stale marks under refold surface a bare NAV.** An execution-time-early observation lands; projections refold the quantities but re-entered marks go stale (VM-7); NAV computes fresh quantities $\times$ stale marks and surfaces a bare number — C-8.1’s NAV type carries no staleness flag. Ties C4.
- **W2 [FL, HIGH; with TA]. New invariants have no firing witness, and no liveness.** Gate 1/2, VM-6, VM-7 map to no property test (CLAUDE.md §3: “zero firings is a defect”). [FL] Gate 2 passes *vacuously* where thin history never generates its precondition; a loose VM-6 bound marks every move “explained.” [TA] detect-and-flag has no liveness — a bulk retraction stale-flags much of the book (MD-8 propagates forward) with no deadline or owner.
- **W3 [TA, HIGH]. Gate 2 blinds risk where it is most needed.** Thin-history names (IPO, post-restructuring) yield permanent “undecidable” — no risk report for the riskiest names; under regime shift (2008/COVID/dividend cut) unprecedented states are rejected as “unrealistic,” suppressing the tail. *Divergence preserved:* TA reads thin history as

runtime *refusal*, FL (W2) as the test’s *vacuous pass* — the refuse-behaviour must be *shown to fire*.

- **W4 [TA, HIGH; with FL]. σ_{prod} poles force structural chain-breaks the alarm cannot distinguish.** At spot near $\Gamma_{\text{adj}} = 0$ an ordinary vanilla reconciles at no real volatility, residual forced $O(1)$, VM-7 calls the chain “broken” through no data fault. *Divergence preserved:* FL reads PE-6’s pole treatment as an honest diagnostic; TA reads the clash with VM-7’s “forbidden” as alarm fatigue — reconciled by exempting degenerate-carry poles from the broken-chain class.
- **W5 [TA, HIGH]. Write amplification meets the single writer and $O(n)$ replay.** VM-1 re-enters every model-priced mark; VM-10/MD-11 re-enter every mark of every scenario and MC path to be replayable. A real book funnels the whole valuation/risk firehose through one writer, and time-travel folds a prefix growing at pricing cadence, not event cadence. No checkpoint is fixed.
- **W6 [FL, MED]. Two correction species collide at the TA-union seam.** A vendor “correction” that also restates the timestamp is both an order-correction (reorders the fold, C-2.7) and a value-correction (MD-10); never distinguished — undefined behaviour.
- **W7 [MK, MED]. Idempotence is conditional on an unnamed assumption.** C-11.5 lists it with no caveat; MD-1 admits two distinct prints identical under every field collide and one is silently dropped. Name the identifier-grain trust assumption beside the invariant.

Part II — Temporal as Execution Substrate for the Whole Framework

The Temporal Committee’s unanimous consensus, assembled at round 11 and signed by all five members (TEMPORAL-1...5) and both referees (FORMALIS, TuringAward). This is a substrate proposal, not a Constitutional amendment: the immutable log remains the sole book of record, and Temporal workflow histories are never the source of truth.

1. Thesis and mapping

Central result. Market data and valuation are *not* a second substrate. The seed (`temporalv16.tex`) mapped the ledger’s two untrusted machines — the Event Monitor and the Events Executor — onto Temporal, keeping the Transaction Executor outside as the single writer. The extension adds no new writer, no new door, no new book: **the whole framework is the one machine folding observations and re-entered observations through the one door.** Every market-data fact, model number, gate verdict, and valuation mark crosses the same door under the same cause-derived identity; everything else is a projection recomputed from the log.

Temporal primitive	Ledger / market-data / valuation object
Workflow (long-lived, keyed <code>unitId</code>)	one unit walking its product graph; pending timers + signal-waits = the current node’s armed watch set (R-03). Derived-object builds, MD-16 dynamic applications, and valuation-chain links are sequences it runs — no new workflow kind.
Activity	every side effect: capture an arrival, read a cut , run a model (version-pinned), propose to the door, apply the market data operator , compute a certificate, evaluate an MD-16 gate. NAV/PnL/operator/settlement are read-time projection activities. A read that records nothing is the forbidden bare read .
Signal / signal-with-start	timing and reference only (a cause identifier, a log position), never economic payload: a watch satisfaction, a corporate-action cascade fan-out, or a “re-read: the fold moved” notice. Ledger-created units (the settlement-obligation unit, a market-claim leg, split successor legs) start their own workflows; graph branches stay in-workflow.
Query	a non-authoritative liveness view; a valuation or collateral dispute is a <i>replay</i> — a read-only re-derivation from the log, never a second authority.
Child workflow	<i>not</i> used for lineage coupling (parental linkage is ledger-absent relational state); a corporate-action cascade fans out by signal-with-start. Child workflows appear only <i>inside a simulation namespace</i> for history-bounding — not the forbidden use.
Continue-as-new	history-size control; carries <code>{unitId lineageId, nodeId, cut}</code> only. The valuation chain , the derived states, and the gate verdicts live on the log and rehydrate from it.
Durable timer / Schedule	timer = a date- watch liveness half only, authoring none of the three times; Schedule = system cadence only (end-of-day sweeps, overdue-watch reconciliation), never a per-unit contractual date.
The Transaction Executor + the immutable log	the trusted single writer, fronted by one idempotent door activity — the only write path. Never a workflow.
The two MD-16 gates, the CA sandwich, the backtest	the gates = version-pinned predicate activities whose verdict + state cross the door in one transaction; the sandwich = a sequence of idempotent legs struck as a projection from the log; a backtest = a strategy unit run through a trajectory in an isolated namespace against its own lineage’s door.

Acceptance test. Wipe the Temporal cluster: the entire workflow population re-derives from the log — unit status re-read, watches re-armed, chains and derived states and gate verdicts rebuilt, nothing lost.

2. The three record kinds and the one lineage discipline

Every object the framework records is exactly one of three kinds, distinguished by **staleness semantics**:

1. **Projection** — computes over the record and stores nothing; recomputed on every read; *never stale*. Balances, the three homes, NAV, profit-and-loss, the valuation chain as assembled, the market data operator, settlement instructions.
2. **Re-entered observation** — a model output evaluated *outside* the fold (the ledger runs no model) whose result re-enters through the one door as a recorded observation; read-back is unconditional;

stale when a consumed input moves (MD-8), superseded forward. A fitted surface, a mark, a greek, a calibration.

3. **Recorded decision, pinned as-known** — a verdict recorded with the event it decides; *not recompute-on-read, not stale-on-input-move* — it decided against inputs as they then stood. This is **not a new primitive**: the base ledger already has it as the door’s admit/refuse; the MD-16 gate verdict is the same shape.

The lineage discipline. Every re-entry carries a cause-derived transaction identifier over the **canonical 3-tuple key** (input-cut, model-version, recipe/dynamic-version). The input-cut is recorded at **exact grain** (a log-position or content-hash, never a coarse “latest” label). A stochastic recipe’s drawn **seed** is a recorded parameter *of the recipe/dynamic-version term* — so the key stays a 3-tuple; a separate seed slot would mint two identifiers for two retries and double-admit, the same failure that keeping the numerical environment in the key would cause. **Seed and numerical-environment are out of identity** — both are recorded Tier-2 re-derivation terms, never txid coordinates. The refold that honours a late arrival is **forward-only** and is the single writer’s work; Temporal is signalled to re-read and re-fires forward, never rewinding its own history against a new fold.

3. Decomposition per framework area

3.1 Ledger. The unit-workflow holds timers, signal-waits, selectors, and sequencing, and nothing else; its in-memory node is a mirror, rehydrated from the ledger on start, on continue-as-new, and on any doubt. Side effects are activities: capture-arrival → evaluate-contract (version-pinned, queue-routed, never a local activity) → propose-to-door. The **Transaction Executor** is an external single-writer service fronted by one idempotent **admit** activity whose write is an **atomic unique-key insert on the cause-derived identifier** — not check-then-append. Exactly-once admission is therefore a **total function of the durable log**: dedup is decided against the committed identifier-set on the log, never an in-memory in-flight set or arrival order. A corporate-action cascade fans out by signal-with-start; each firing proposes its own identifier, so all n legs commit. Settlement is a pure projection that writes nothing; settlement failure re-enters as a recorded event walking the settlement-obligation unit’s graph — never a Temporal compensation or saga.

3.2 Market data. Capture records the observation through the **one door** (envelope-first, before payload validation); absorption of a redelivery is the door’s cause-derived-identifier job at the registered grain. A **projection-kind** derived object (a discount factor, an operator-adjusted value) is a read-time activity that stores nothing. A **re-entered-observation-kind** derived object (a fitted surface, a filter) runs the model in a version-pinned activity whose seed is recorded to the log before compute and reused on resume, then re-enters through the door.

The two MD-16 gates and the single-transaction write. Applying a declared *dynamic* to an admissible base state yields a derived state, admitted only through two gates decided at application time: **Gate 1**, no-arbitrage by construction ($m^* \in \Theta_{AF}$); **Gate 2**, realism against the underlying’s own as-known history (per-functional percentiles, joint where marginals would hide an inversion). The workflow pins **one cut** and runs **gate1**, **gate2**, and **construct** all over that cut (MD-12), then **the derived state and its passing gate verdict cross the door as one transaction** — never a pass recorded first with construction gated on it, which would open an inconsistent-read window. Fail or undecidable is a recorded **refusal**; the state is never constructed, and a consumer names a state only by its admission record, so an ungated state is unnameable.

Prevention-at-construction = detection-at-audit. Gate 1 is a decidable predicate, but one requiring model knowledge (the arbitrage-free set) the door must not hold — the door is generic machinery with no product knowledge. So the gate runs in a version-pinned activity, like a contract; the door commits verdict-and-state atomically; a mis-gate degrades to detection-at-audit by recomputation, exactly as an economically-wrong contract does. Under the single-trusted-door model, that is the honest status of “prevention” here.

$A' = \mathbf{FLAG}$. A correction to a consumed input that lands under the pinned cut before admission does **not** refuse: it **flags m^* stale-forward** (MD-8/MD-10), m^* remaining the as-known-at-cut value it was gated as — the single writer decides staleness on the refold. The state is kind-2 (staleable), the decision kind-3 (pinned, never stale); refusing would conflate them, and C-11.3 is a structural consistency guard, not a freshness check. **Refuse is reserved for a gate fail/undecidable verdict (the gate decides) or an unresolvable structural reference (the door decides) — never a fresher tip.** A 90-second calibration under corrections every 45 seconds never converges under REFUSE; FLAG guarantees progress.

The value-level β bound. A model number is one member of a set the attempts could have produced; read-back reproduces the recorded bytes, not proof the mark is within tolerance of an honest re-derivation. The producer attests a **reproducibility class** carrying a bound β in the lineage. The **door checks**

presence and structure only: it never compares β to a tolerance, since the tolerance is the *consuming* instrument's and is not single-valued at admission (one surface at $\beta = 3$ bp serves a unit tolerant to 5 bp and one to 1 bp). A missing attestation is admit-and-flag, never refused. **At consumption, $\beta \leq \text{VM-6}$ is checked as a VM-7 broken chain:** the consuming unit's valuation projection raises a broken chain when β exceeds *that* unit's tolerance, or is absent. Totality is by type (**COVERAGE- β**): the sole accessor to a kind-2 leaf is the "current fit" selector, which carries the class and raises the chain, so a raw path is a type error. A false β degrades to detection-at-audit.

The compute/emit split removes the door-arrival race for a non-bit-reproducible model: **runModel** is one activity whose output Temporal memoizes; **proposeToDoor** is a separate activity whose only input is that recorded output — a typed boundary, so a worker that recomputes-and-proposes in one step (which would reopen the race) is not representable. Retries re-present identical bytes; canonical-by-first is a declared rule, and bit-reproducibility is never an admission precondition (out-of-scope numerics).

Corporate actions. A corporate action is a first-class recorded transaction; the market data operator is a read-time projection, the original never overwritten. A late or corrected corporate action is a reordering handled by the refold, never a Temporal un-adjust.

3.3 Valuation. A **chain link** reads a cut, prices a leg (version-pinned, re-entering via the door), computes the **profit-and-loss explain certificate** with entry-and-exit greeks, checks the residual against its declared bound, and appends the link through the door. The chain and its certificates live on the log; the workflow holds only the chain-head cursor. NAV and profit-and-loss are pure projections — one recipe also serves risk, so risk and the books cannot disagree by construction.

Re-mark cadence splits by the spec's own axis. End-of-day desk marks are system cadence → a **Schedule sweep** that chooses which units to re-mark, the pricing activity reading each unit's node/frame/cut *from the record*, so a mid-corporate-action unit is never priced in a stale frame. Contractual, corporate-action-driven, or input-moved re-marks are a **per-unit watch** reading the node.

The corporate-action valuation sandwich is a valuation struck immediately before the transition and another immediately after, the market data operator transporting the pre-frame mark into the post-frame as a zero-profit identity, with a full explain across the pair. It is a **sequence of idempotent legs** whose completion is a deterministic function of the recorded {nodeId, cut}: a mark is kind-1 when it is the operator's frame re-coordination, kind-2 when model-priced; the certificate is the one kind-3 admission. Because no leg is held in workflow memory, a continue-as-new at any step re-derives the whole sandwich or resumes the one missing certificate — **a half-sandwich is never nameable**. The residual within its three declared tolerances (minor-unit rounding, the convention's rounding, any re-derived surface's calibration tolerance) is the proof the action went well.

Risk, scenarios, and backtests run the one per-unit recipe on shifted market data in an isolated namespace against its own lineage's door. The **shift** is the recorded seed; a strategy is itself a **unit**, so a backtest reuses one-workflow-per-unit unchanged; simulated re-entries enter the simulated path's own record, and where coordinates coincide with production the chain equals it by determinism, reached by recomputation, never a second stored copy.

4. Invariants, the canonical key, and the production/simulation seam

- **I1 — one single writer per lineage.** The write credential is a **fenced lease over a quorum-committed log**: an append is acknowledged only after it is durable in the quorum (durable-before-ack, so no lost admission), and the log rejects any append not carrying the current fence token, so a split-brain stale door's writes are rejected by the log — never two admitting doors. Failover moves the fence, never duplicates it.
- **I2 — the identity key.** The canonical 3-tuple (input-cut, model-version, recipe/dynamic-version); seed and numerical-environment out of identity; input-cut exact-grained. The injectivity dual: distinct fine-grained causes yield distinct admissions.
- **I3 — no model in-fold.** The refold is apply $\circ$ contract, a pure fold; kind-2 outputs are immutable leaves; re-derivation is forward and out-of-fold. Enforced by type.
- **I4 — axis non-leak.** The activity reads the recipe/model/dynamic version *from the log*, never from its binary; the orchestration Build-ID pins orchestration only and is not in the fold. Three economics axes (ProductTerms, model/recipe, dynamic/gate-terms) live on the log.

The production/simulation seam is two orthogonal separations. **Base vs derived stream** is a *stream* boundary within one production lineage — tagged sub-streams sharing the production door; a production-serving derived object (today's calibrated surface serving a real mark) is production data. **Production vs simulation** is a *namespace* boundary: a scenario, backtest, or authorised fork runs in an isolated namespace against its own lineage's door, and only a final result re-enters production as an observation. Door credentials follow the second boundary: the production door holds the sole production

write credential, a simulation namespace's door a separate one. The discriminating test: *does a real production unit's valuation chain read it back?*

5. Divergence catalogue and red-team survival

Divergences (D1–D16), each with its containment.

- **D1 Bare-read** — a fetch that records nothing is forbidden; every ingestion activity's only success is proposing an observation-recording transaction, and read activities read the record, never a live feed.
- **D2 Model nondeterminism** — model runs are version-pinned activities (never workflow, never local), output recorded; heavy runs heartbeat under a bounded schedule-to-close so a retry fires only on genuine failure.
- **D3 Past-dated synthesised firings** — the refold's past-dated synthesis is the single writer's work; the substrate re-fires forward only; the late corporate-action sandwich rides this exact path.
- **D4 Staleness vs workflow state** — staleness is a recorded fact; the “current fit” projection selects the latest non-superseded value and carries the flag; orchestration only triggers forward re-derivation.
- **D5 History size** — marks, states, and verdicts live on the log; continue-as-new carries the identifier triple only; derived-state volume is isolated to simulation namespaces.
- **D6 Undecidable / refuse vs retry-to-infinity** — undecidable, refuse, and a broken chain are returned values, not retryable errors; each stands as a visible blocked or overdue item.
- **D7 Prevention in an untrusted constructor** — a decidable predicate any party recomputes; prevention-at-construction plus consumption-by-reference, detection-at-audit; no door privilege added.
- **D8 Workflow-code nondeterminism** — wall clock, randomness, unordered-map iteration, and direct I/O are confined to activities; the three times come from the log.
- **D9 Versioning — three axes** — orchestration Build-ID; contract economics; model/recipe/dynamic/gate terms on the log, never touching workflow code.
- **D10 Retry vs exactly-once admission** — the cause-derived identifier over the canonical 3-tuple, never a Temporal run/attempt id; dedup is load-bearing by construction, and only the pre-log early-drop is optimisation.
- **D11/D12 Queue/timer time vs the fold** — the committed order is the total order (execution, door, hash) at the door alone; timers carry no time authority; a late observation refolds; a late-firing timer yields the identical transaction.
- **D13 Attribution/dispersion convention as worker config** — the convention (the held-invariant factor, the partition, the dispersion rule, the normalisation) is a declared, recorded term read by the projection, never a worker default.
- **D14 Admission $\neq$ deterministic value** — closed by the compute/emit split, canonical-by-first, and the β bound at consumption.
- **D15 MD-16 write atomicity** — state + verdict as one transaction over one pinned cut; $A' = \text{FLAG}$.
- **D16 Value spread bound** — the producer-attested reproducibility class, $\beta \leq \text{VM-6}$ at consumption (COVERAGE- β), false β to audit.

The seven red-team scenarios (S1–S7), and the property each demonstrated survives.

- **S4 Exactly-once = a total function of the durable log** (the root; S1/S3/S6/S7 reduce to it). The atomic unique-key insert admits exactly one row per identifier under N redeliveries $\times$ W racing workers $\times$ a door crash-restart; none is starved; differing payloads under one identifier collapse to first-wins, β bounding the discard.
- **S1 Refold-atomic.** Append-only $\Rightarrow$ rewind not representable; no model recomputed in-fold; re-proposed transactions and synthesised firings collapse to one row; the resume key is the stable (exec, door, hash) triple, never an ordinal. This is what $A' = \text{FLAG}$ rests on.
- **S7 Gate-state-atomic.** State + verdict are one transaction over one pinned cut; a failover re-runs the gate and construct as pure functions of the cut to the same identifier; no ungated or half-verdict state is nameable.
- **S3 Sandwich-carries-no-workflow-state.** The sandwich holds no economic state in Temporal history; a continue-as-new re-derives it or resumes the one missing certificate; a pre-admitted leg is byte-identical by read-back, a re-driven leg is idempotent and β -bounded.
- **S6 Log-is-sole-truth.** Rebuild reads only the log; the cache holds no write credential; a replayed identifier is absorbed; a fabricated novel identifier is door-refused, and a value-corrupted-but-consistent one is audit-caught. *Honest edge, no overclaim:* economic causality is **detection-at-audit, not**

door-prevention — the guarantee is “no poison silently becomes trusted truth; no rebuilt state is not derivable from the log,” not “no structurally-valid poison ever touches the log.”

- **S2 Deploy-is-orchestration-only.** A deploy touches only the Build-ID, which is not in the fold; every value is a fold/read-back under recorded economic versions read from the log; an economic change is a new recipe-version and a new identifier, never a silent rewrite.
- **S5 Three-times-are-recorded-values.** The fold orders on the door’s logical (exec, door, hash) read from the log; a bare clock-read stamping a recorded fact is unrepresentable by type; skew enters only timer firing, so a skewed fire yields the identical transaction and degrades to an overdue-watch event.

6. The firing-witness harvest

A guarantee defended only in prose is not an invariant. Each property below is owed to the property-test regime over generated products, events, and histories, and must be shown to **fire** — a precondition never generated is a defect, not a green test.

Property	Generator that witnesses it fire	Obligation
<code>prop_refoldIdempotent</code>	random late-arrival interleavings + a second refold pass	refold twice = refold once
<code>prop_refoldEqualsTimely</code>	late arrivals vs the same events folded in execution order	refolded state = timely state
<code>prop_exactlyOnceAdmission</code> (S4, double-admit half of I2)	N redeliveries $\times$ W workers $\times$ a door crash-restart, same identifier	exactly one row per identifier; none starved
<code>prop_noSilentUnderAdmit</code> (S4, injectivity half of I2)	two fine-grained-distinct causes with near-identical input-cuts	distinct causes $\rightarrow$ two distinct admitted identifiers
<code>prop_gateStateAtomic</code> (S7)	failover at each of gate1 / gate2 / construct / propose	whole {state, verdict} pair or nothing
<code>prop_sandwichCANInvariant</code> (S3)	continue-as-new after a leg’s emission is durable, and mid-compute of a kind-2 leg	pre-admitted leg byte-identical by read-back; a re-driven leg idempotent, no half/no double, β -bounded
<code>prop_wipeRebuildEqualsLog</code> (S6)	wipe Temporal and rebuild from the log	rebuilt state = pre-wipe state, cache-independent
<code>prop_fabricatedTxidRefused</code> (S6)	On-fabrications: a novel/inconsistent identifier, and a structurally-valid, value-corrupted one (real logged cause, wrong value)	the first door-refused; the second audit-caught (D7) — the audit branch fires
<code>prop_deployMidBacktestInvariant</code> (S2)	Build-ID change at each backtest step	admitted (txid, value) set + order unchanged
<code>prop_clockSkewInvariant</code> (S5)	adversarial per-worker/DC offsets over a history with a commuting same-execution-time pair	admitted set, fold result, execution+monitor times invariant; door-order invariant up to a same-execution-time swap; a skewed timer fire yields the identical identifier
<code>prop_everyKind2ConsumerChecksBeta</code> (COVERAGE- β)	all paths with β above/below tolerance	every $\beta >$ tolerance path raises a VM-7 broken chain

Parking test exercised, empty. The derived stream is not a second store (one immutable-log mechanism on a tagged sub-stream); a gate verdict is a pinned event-outcome, not a recompute-on-read projection; the late-corporate-action sandwich is the reordering path, not a compensation. The design must **not** turn on the Valuation Manifesto’s PARK-1, which the MD-16 mechanism neither reopens nor turns on. Residual open items — the truthfulness of a producer’s attested β (a perimeter reconciliation caught at audit), and the load model sizing the continue-as-new cadence and the door and derivation pools — are parked, not swept.

7. Minority report

None. Unanimous consensus at round 11: TEMPORAL-1 through TEMPORAL-5 all SIGN, and both referees (FORMALIS, TuringAward) sign the round-11 text. The design is Pareto-optimal across correctness, minimalism, and simplicity, and — with `prop_noSilentUnderAdmit` added and the harvest completed — across testability. No entry demands anything of the Constitution; every object stays a projection, a re-entered observation, or a recorded decision on the log, and the log stays the sole book of record.